favorite principles—that theoretical physics should be "radically conservative." The idea is that a successful physical theory is one that has been tested against experimental data, but only in regimes that experimenters are actually able to reach. One should be conservative, in the sense that we should start with the theories and principles that are already established as successful, rather than arbitrarily introducing new approaches whenever new phenomena are encountered. But one should also be radical, in the sense that the predictions and implications of our theories should be taken seriously in regimes well outside where they have been tested. The phrases "we should start" and "should be taken seriously" are crucial here; of course new theories are warranted when old ones are shown to blatantly contradict the data, and just because a prediction is taken seriously doesn't mean it shouldn't be revised in light of new information. But Wheeler's philosophy was that we should start prudently, with aspects of nature we believe we understand, and then act boldly, extrapolating our best ideas to the ends of the universe.

Part of Everett's inspiration was the search for a theory of quantum gravity, which Wheeler had recently become interested in. The rest of physics—matter, electromagnetism, the nuclear forces—seems to fit comfortably within the framework of quantum mechanics. But gravity was (and remains) a stubborn exception. In 1915, Einstein proposed the general theory of relativity, according to which spacetime itself is a dynamical entity whose bends and warps are what you and I perceive as the force of gravity. But general relativity is a thoroughly classical theory, with analogues of position and momentum for the curvature of space-time, and no limits on how we might measure them. Taking that theory and "quantizing" it, constructing a theory of wave functions of space-time rather than particular classical spacetimes has proven difficult.